

Chapter 0

A short mathematical review

A basic understanding of pre-calculus, calculus, and complex numbers is required for this course. This zero chapter presents a concise review.

0.1 The trigonometric functions

The Pythagorean trigonometric identity is

$$\sin^2 x + \cos^2 x = 1,$$

and the addition theorems are

$$\begin{aligned}\sin(x+y) &= \sin(x)\cos(y) + \cos(x)\sin(y), \\ \cos(x+y) &= \cos(x)\cos(y) - \sin(x)\sin(y).\end{aligned}$$

Also, the values of $\sin x$ in the first quadrant can be remembered by the rule of quarters, with $0^\circ = 0$, $30^\circ = \pi/6$, $45^\circ = \pi/4$, $60^\circ = \pi/3$, $90^\circ = \pi/2$:

$$\begin{aligned}\sin 0^\circ &= \sqrt{\frac{0}{4}}, & \sin 30^\circ &= \sqrt{\frac{1}{4}}, & \sin 45^\circ &= \sqrt{\frac{2}{4}}, \\ \sin 60^\circ &= \sqrt{\frac{3}{4}}, & \sin 90^\circ &= \sqrt{\frac{4}{4}}.\end{aligned}$$

The following symmetry properties are also useful:

$$\sin(\pi/2 - x) = \cos x, \quad \cos(\pi/2 - x) = \sin x;$$

and

$$\sin(-x) = -\sin(x), \quad \cos(-x) = \cos(x).$$

0.2 The exponential function and the natural logarithm

The transcendental number e , approximately 2.71828, is defined as

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n.$$

The exponential function $\exp(x) = e^x$ and natural logarithm $\ln x$ are inverse functions satisfying

$$e^{\ln x} = x, \quad \ln e^x = x.$$

The usual rules of exponents apply:

$$e^x e^y = e^{x+y}, \quad e^x / e^y = e^{x-y}, \quad (e^x)^p = e^{px}.$$

The corresponding rules for the logarithmic function are

$$\ln(xy) = \ln x + \ln y, \quad \ln(x/y) = \ln x - \ln y, \quad \ln x^p = p \ln x.$$

0.3 Definition of the derivative

The derivative of the function $y = f(x)$, denoted as $f'(x)$ or dy/dx , is defined as the slope of the tangent line to the curve $y = f(x)$ at the point (x, y) . This slope is obtained by a limit, and is defined as

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}. \quad (1)$$

0.4 Differentiating a combination of functions

0.4.1 The sum or difference rule

The derivative of the sum of $f(x)$ and $g(x)$ is

$$(f + g)' = f' + g'.$$

Similarly, the derivative of the difference is

$$(f - g)' = f' - g'.$$

0.4.2 The product rule

The derivative of the product of $f(x)$ and $g(x)$ is

$$(fg)' = f'g + fg',$$

and should be memorized as “the derivative of the first times the second plus the first times the derivative of the second.”

0.4.3 The quotient rule

The derivative of the quotient of $f(x)$ and $g(x)$ is

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2},$$

and should be memorized as “the derivative of the top times the bottom minus the top times the derivative of the bottom over the bottom squared.”

0.4.4 The chain rule

The derivative of the composition of $f(x)$ and $g(x)$ is

$$\left(f(g(x))\right)' = f'(g(x)) \cdot g'(x),$$

and should be memorized as “the derivative of the outside times the derivative of the inside.”

0.5 Differentiating elementary functions

0.5.1 The power rule

The derivative of a power of x is given by

$$\frac{d}{dx}x^p = px^{p-1}.$$

0.5.2 Trigonometric functions

The derivatives of $\sin x$ and $\cos x$ are

$$(\sin x)' = \cos x, \quad (\cos x)' = -\sin x.$$

We thus say that “the derivative of sine is cosine,” and “the derivative of cosine is minus sine.” Notice that the second derivatives satisfy

$$(\sin x)'' = -\sin x, \quad (\cos x)'' = -\cos x.$$

0.5.3 Exponential and natural logarithm functions

The derivative of e^x and $\ln x$ are

$$(e^x)' = e^x, \quad (\ln x)' = \frac{1}{x}.$$

0.6 Definition of the integral

The definite integral of a function $f(x) > 0$ from $x = a$ to b ($b > a$) is defined as the area bounded by the vertical lines $x = a$, $x = b$, the x -axis and the curve $y = f(x)$. This “area under the curve” is obtained by a limit. First, the area is approximated by a sum of rectangle areas. Second, the integral is defined to be the limit of the rectangle areas as the width of each individual rectangle goes to zero and the number of rectangles goes to infinity. This resulting infinite sum is called a *Riemann Sum*, and we define

$$\int_a^b f(x)dx = \lim_{h \rightarrow 0} \sum_{n=1}^N f(a + (n-1)h) \cdot h, \quad (2)$$

where $N = (b - a)/h$ is the number of terms in the sum. The symbols on the left-hand-side of (2) are read as “the integral from a to b of f of x dee x .” The Riemann Sum definition is extended to all values of a and b and for all values of $f(x)$ (positive and negative). Accordingly,

$$\int_b^a f(x)dx = -\int_a^b f(x)dx \quad \text{and} \quad \int_a^b (-f(x))dx = -\int_a^b f(x)dx.$$

Also,

$$\int_a^c f(x)dx = \int_a^b f(x)dx + \int_b^c f(x)dx,$$

which states when $f(x) > 0$ and $a < b < c$ that the total area is equal to the sum of its parts.

0.7 The fundamental theorem of calculus

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Using the definition of the derivative, we differentiate the following integral:

$$\begin{aligned}\frac{d}{dx} \int_a^x f(s) ds &= \lim_{h \rightarrow 0} \frac{\int_a^{x+h} f(s) ds - \int_a^x f(s) ds}{h} \\ &= \lim_{h \rightarrow 0} \frac{\int_x^{x+h} f(s) ds}{h} \\ &= \lim_{h \rightarrow 0} \frac{hf(x)}{h} \\ &= f(x).\end{aligned}$$

This result is called the fundamental theorem of calculus, and provides a connection between differentiation and integration.

The fundamental theorem teaches us how to integrate functions. Let $F(x)$ be a function such that $F'(x) = f(x)$. We say that $F(x)$ is an antiderivative of $f(x)$. Then from the fundamental theorem and the fact that the derivative of a constant equals zero,

$$F(x) = \int_a^x f(s) ds + c.$$

Now, $F(a) = c$ and $F(b) = \int_a^b f(s) ds + F(a)$. Therefore, the fundamental theorem shows us how to integrate a function $f(x)$ provided we can find its antiderivative:

$$\int_a^b f(s) ds = F(b) - F(a). \quad (3)$$

Unfortunately, finding antiderivatives is much harder than finding derivatives, and indeed, most complicated functions cannot be integrated analytically.

We can also derive the very important result (3) directly from the definition of the derivative (1) and the definite integral (2). We will see it is convenient to choose the same h in both limits. With $F'(x) = f(x)$, we have

$$\begin{aligned}\int_a^b f(s) ds &= \int_a^b F'(s) ds \\ &= \lim_{h \rightarrow 0} \sum_{n=1}^N F'(a + (n-1)h) \cdot h \\ &= \lim_{h \rightarrow 0} \sum_{n=1}^N \frac{F(a + nh) - F(a + (n-1)h)}{h} \cdot h \\ &= \lim_{h \rightarrow 0} \sum_{n=1}^N F(a + nh) - F(a + (n-1)h).\end{aligned}$$

The last expression has an interesting structure. All the values of $F(x)$ evaluated at the points lying between the endpoints a and b cancel each other in consecutive terms. Only the value $-F(a)$ survives when $n = 1$, and the value $+F(b)$ when $n = N$, yielding again (3).

0.8 Definite and indefinite integrals

The Riemann sum definition of an integral is called a *definite integral*. It is convenient to also define an indefinite integral by

$$\int f(x)dx = F(x),$$

where $F(x)$ is the antiderivative of $f(x)$.

0.9 Indefinite integrals of elementary functions

From our known derivatives of elementary functions, we can determine some simple indefinite integrals. The power rule gives us

$$\int x^n dx = \frac{x^{n+1}}{n+1} + c, \quad n \neq -1.$$

When $n = -1$, and x is positive, we have

$$\int \frac{1}{x} dx = \ln x + c.$$

If x is negative, using the chain rule we have

$$\frac{d}{dx} \ln(-x) = \frac{1}{x}.$$

Therefore, since

$$|x| = \begin{cases} -x & \text{if } x < 0; \\ x & \text{if } x > 0, \end{cases}$$

we can generalize our indefinite integral to strictly positive or strictly negative x :

$$\int \frac{1}{x} dx = \ln|x| + c.$$

Trigonometric functions can also be integrated:

$$\int \cos x dx = \sin x + c, \quad \int \sin x dx = -\cos x + c.$$

Easily proved identities are an addition rule:

$$\int (f(x) + g(x))dx = \int f(x)dx + \int g(x)dx;$$

and multiplication by a constant:

$$\int Af(x)dx = A \int f(x)dx.$$

This permits integration of functions such as

$$\int (x^2 + 7x + 2)dx = \frac{x^3}{3} + \frac{7x^2}{2} + 2x + c,$$

and

$$\int (5 \cos x + \sin x)dx = 5 \sin x - \cos x + c.$$

0.10 Substitution

More complicated functions can be integrated using the chain rule. Since

$$\frac{d}{dx}f(g(x)) = f'(g(x)) \cdot g'(x),$$

we have

$$\int f'(g(x)) \cdot g'(x) dx = f(g(x)) + c.$$

This integration formula is usually implemented by letting $y = g(x)$. Then one writes $dy = g'(x)dx$ to obtain

$$\begin{aligned} \int f'(g(x))g'(x)dx &= \int f'(y)dy \\ &= f(y) + c \\ &= f(g(x)) + c. \end{aligned}$$

0.11 Integration by parts

Another integration technique makes use of the product rule for differentiation. Since

$$(fg)' = f'g + fg',$$

we have

$$f'g = (fg)' - fg'.$$

Therefore,

$$\int f'(x)g(x)dx = f(x)g(x) - \int f(x)g'(x)dx.$$

Commonly, the above integral is done by writing

$$\begin{aligned} u &= g(x) & dv &= f'(x)dx \\ du &= g'(x)dx & v &= f(x). \end{aligned}$$

Then, the formula to be memorized is

$$\int u dv = uv - \int v du.$$

0.12 Taylor series

A Taylor series of a function $f(x)$ about a point $x = a$ is a power series representation of $f(x)$ developed so that all the derivatives of $f(x)$ at a match all the derivatives of the power series. Without worrying about convergence here, we have

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \dots$$

Notice that the first term in the power series matches $f(a)$, all other terms vanishing, the second term matches $f'(a)$, all other terms vanishing, etc. Commonly, the Taylor

series is developed with $a = 0$. We will also make use of the Taylor series in a slightly different form, with $x = x_* + \epsilon$ and $a = x_*$:

$$f(x_* + \epsilon) = f(x_*) + f'(x_*)\epsilon + \frac{f''(x_*)}{2!}\epsilon^2 + \frac{f'''(x_*)}{3!}\epsilon^3 + \dots$$

Another way to view this series is that of $g(\epsilon) = f(x_* + \epsilon)$, expanded about $\epsilon = 0$. Taylor series that are commonly used include

$$\begin{aligned} e^x &= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots, \\ \sin x &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots, \\ \cos x &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots, \\ \frac{1}{1+x} &= 1 - x + x^2 - \dots, \quad \text{for } |x| < 1, \\ \ln(1+x) &= x - \frac{x^2}{2} + \frac{x^3}{3} - \dots, \quad \text{for } |x| < 1. \end{aligned}$$

0.13 Functions of several variables

For simplicity, we consider a function $f = f(x, y)$ of two variables, though the results are easily generalized. The partial derivative of f with respect to x is defined as

$$\frac{\partial f}{\partial x} = \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h},$$

and similarly for the partial derivative of f with respect to y . To take the partial derivative of f with respect to x , say, take the derivative of f with respect to x holding y fixed. As an example, consider

$$f(x, y) = 2x^3y^2 + y^3.$$

We have

$$\frac{\partial f}{\partial x} = 6x^2y^2, \quad \frac{\partial f}{\partial y} = 4x^3y + 3y^2.$$

Second derivatives are defined as the derivatives of the first derivatives, so we have

$$\frac{\partial^2 f}{\partial x^2} = 12xy^2, \quad \frac{\partial^2 f}{\partial y^2} = 4x^3 + 6y;$$

and the mixed second partial derivatives are

$$\frac{\partial^2 f}{\partial x \partial y} = 12x^2y, \quad \frac{\partial^2 f}{\partial y \partial x} = 12x^2y.$$

In general, mixed partial derivatives are independent of the order in which the derivatives are taken.

Partial derivatives are necessary for applying the chain rule. Consider

$$df = f(x + dx, y + dy) - f(x, y).$$

We can write df as

$$\begin{aligned} df &= [f(x+dx, y+dy) - f(x, y+dy)] + [f(x, y+dy) - f(x, y)] \\ &= \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy. \end{aligned}$$

If one has $f = f(x(t), y(t))$, say, then

$$\frac{df}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}.$$

And if one has $f = f(x(r, \theta), y(r, \theta))$, say, then

$$\frac{\partial f}{\partial r} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial r}, \quad \frac{\partial f}{\partial \theta} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial \theta} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial \theta}.$$

A Taylor series of a function of several variables can also be developed. Here, all partial derivatives of $f(x, y)$ at (a, b) match all the partial derivatives of the power series. With the notation

$$f_x = \frac{\partial f}{\partial x}, \quad f_y = \frac{\partial f}{\partial y}, \quad f_{xx} = \frac{\partial^2 f}{\partial x^2}, \quad f_{xy} = \frac{\partial^2 f}{\partial x \partial y}, \quad f_{yy} = \frac{\partial^2 f}{\partial y^2}, \quad \text{etc.},$$

we have

$$\begin{aligned} f(x, y) &= f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b) \\ &+ \frac{1}{2!} \left(f_{xx}(a, b)(x - a)^2 + 2f_{xy}(a, b)(x - a)(y - b) + f_{yy}(a, b)(y - b)^2 \right) + \dots \end{aligned}$$

0.14 Complex numbers

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[View tutorial on YouTube: Complex Exponential Function](#)

We define the imaginary number i to be one of the two numbers that satisfies the rule $(i)^2 = -1$, the other number being $-i$. Formally, we write $i = \sqrt{-1}$. A complex number z is written as

$$z = x + iy,$$

where x and y are real numbers. We call x the real part of z and y the imaginary part and write

$$x = \operatorname{Re} z, \quad y = \operatorname{Im} z.$$

Two complex numbers are equal if and only if their real and imaginary parts are equal.

The complex conjugate of $z = x + iy$, denoted as $\bar{z}$, is defined as

$$\bar{z} = x - iy.$$

Using z and $\bar{z}$, we have

$$\operatorname{Re} z = \frac{1}{2} (z + \bar{z}), \quad \operatorname{Im} z = \frac{1}{2i} (z - \bar{z}). \quad (4)$$